did not enjoy the advantages of either of the two more extreme strategies. The exploiters would have evolved smaller and smaller size, and faster mobility. The honest ones would have evolved larger and larger size, to compensate for the ever-smaller investment contributed by the exploiters, and they became immobile because they would always be actively chased by the exploiters anyway. Each honest one would 'prefer' to fuse with another honest one. But the selection pressure to lock out exploiters would have been weaker than the pressure on exploiters to duck under the barrier: the exploiters had more to lose, and they therefore won the evolutionary battle. The honest ones became eggs, and the exploiters became sperms.

Males, then, seem to be pretty worthless fellows, and on simple 'good of the species' grounds, we might expect that males would become less numerous than females. Since one male can theoretically produce enough sperms to service a harem of 100 females we might suppose that females should outnumber males in animal populations by 100 to 1. Other ways of putting this are that the male is more 'expendable', and the female more 'valuable' to the species. Of course, looked at from the point of view of the species as a whole, this is perfectly true. To take an extreme example, in one study of elephant seals, 4 per cent of the males accounted for 88 per cent of all the copulations observed. In this case, and in many others, there is a large surplus of bachelor males who probably never get a chance to copulate in their whole lives. But these extra males live otherwise normal lives, and they eat up the population's food resources no less hungrily than other adults. From a 'good of the species' point of view this is horribly wasteful; the extra males might be regarded as social parasites. This is just one more example of the difficulties that the group selection theory gets into. The selfish gene theory, on the other hand, has no trouble in explaining the fact that the numbers of males and females tend to be equal, even when the males who actually reproduce may be a small fraction of the total number. The explanation was first offered by R. A. Fisher.

The problem of how many males and how many females are born is a special case of a problem in parental strategy. Just as we discussed the optimal family size for an individual parent trying to maximize her gene survival, we can also discuss the optimal sex ratio. Is it better to entrust your precious genes to sons or to daughters? Suppose a mother invested all her resources in sons, and therefore had none left to invest in